BACKEND ENGINEERING · HLD INTERVIEW PREP

Proxy Servers

System Design — Complete Study Notes

Forward Proxy · Reverse Proxy · VPN · Load Balancer
Firewall · CDN · DDoS Protection · Caching

TOPICS COVERED

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1. What is a Proxy Server?

A **proxy server** is an intermediary between a client and a destination server. Instead of the client communicating directly with the server, it sends requests to the proxy, which forwards them on the client's behalf and returns the response. The destination server only ever sees the proxy — not the real client.

Analogy: A child wants chocolate from a shop. Instead of going directly, the child asks the mother (proxy). The mother buys it and hands it to the child. The shopkeeper only deals with the mother — the child's identity is hidden.

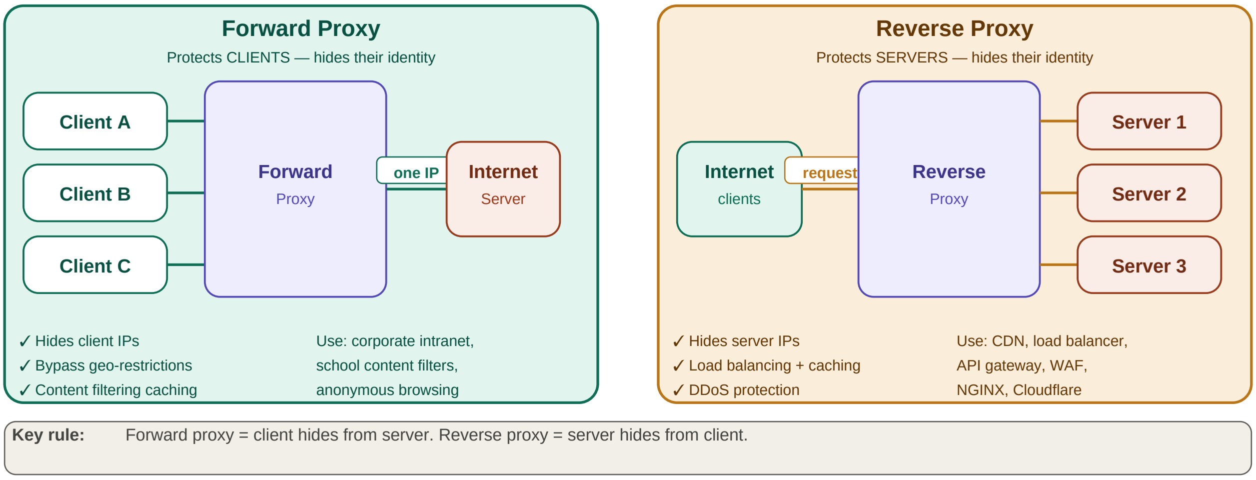
Two directions: Forward proxy protects clients (client side). Reverse proxy protects servers (server side).

All proxies operate at the Application Layer (L7) — they understand HTTP/HTTPS content, not just packets.

Proxies can cache, filter, log, authenticate, and route — far more than a simple IP forwarder.

2. Forward vs Reverse Proxy

Forward Proxy vs Reverse Proxy

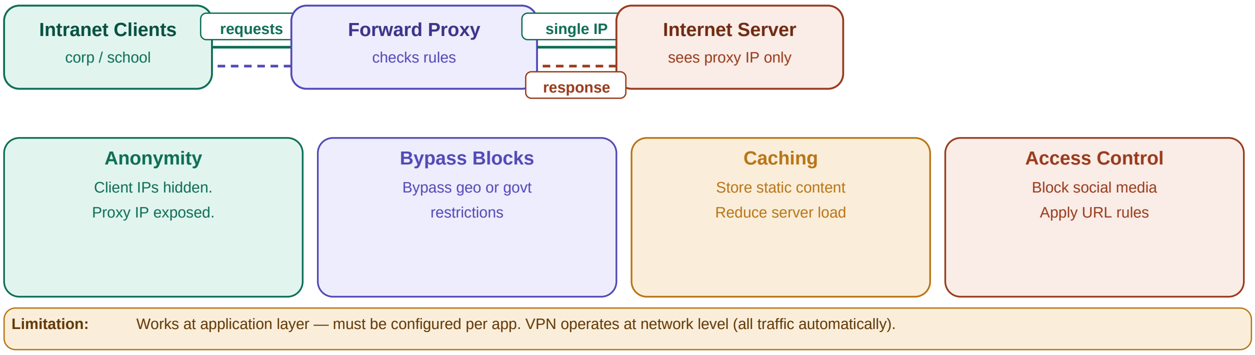


Forward proxy: clients hidden → proxy → server (one IP shown) · Reverse proxy: internet → proxy → multiple backend servers (server IPs hidden)

3. Forward Proxy — Features & Use Cases

A forward proxy sits between internal clients (corporate LAN, school network) and the external internet. Clients know they are using a proxy and configure their applications to route through it.

Forward Proxy — Detailed Flow Features

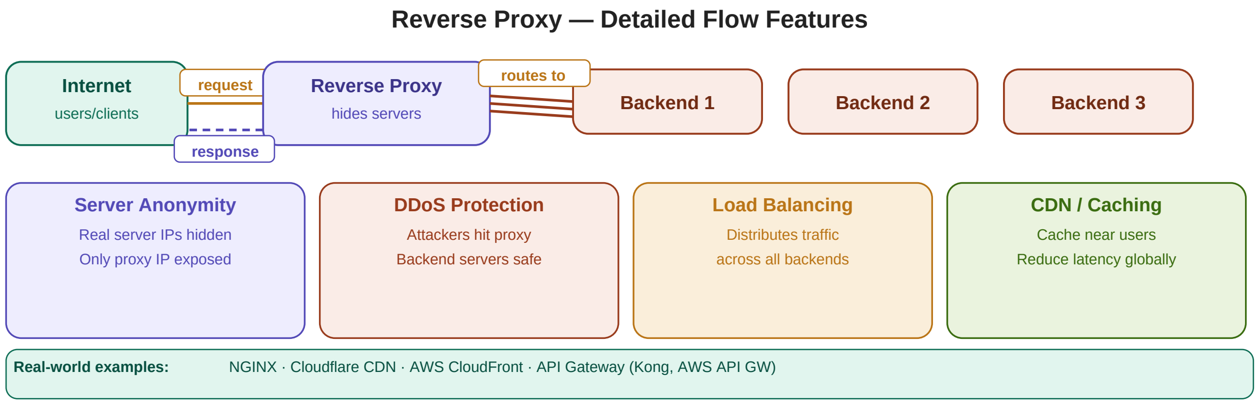


Forward proxy flow: multiple clients → proxy (applies rules) → single IP to server · Features: anonymity, bypass, caching, access control

Feature	How it works	Example
Anonymity	Client IPs replaced by proxy IP	Corporate LAN users browse anonymously
Bypass restrictions	Proxy in another region fetches content	Access geo-blocked video from abroad
Request grouping	Multiple clients making same request served from cache	10 users request same image — one fetch
Access control	Admin configures blocked URL patterns	Block reddit.com at school proxy
Caching	Frequently accessed content stored locally	News site HTML cached — no re-fetch

4. Reverse Proxy — Features & Use Cases

A reverse proxy sits in front of backend servers and intercepts all incoming internet traffic. External users never know which or how many backend servers exist — they only ever interact with the reverse proxy. This is the architecture behind CDNs, NGINX, and cloud load balancers.



Reverse proxy flow: internet → proxy → routes to backend 1/2/3 · Features: anonymity, DDoS protection, load balancing, CDN caching

Feature	How it works	Real-world example
Server anonymity	Only proxy IP visible — backends hidden	NGINX in front of Node.js cluster
DDoS protection	Attackers flood proxy — backends unaffected	Cloudflare absorbs DDoS attack
Load balancing	Round-robin / weighted routing to backends	AWS ALB routes to 10 EC2 instances
CDN caching	Content cached at edge servers near users	Cloudflare caches JS/CSS near Mumbai
SSL termination	Proxy handles HTTPS — backends serve HTTP	NGINX handles TLS, forwards HTTP

5, 6 & 7. Proxy vs VPN vs Load Balancer vs Firewall

Proxy vs VPN

Aspect	Proxy	VPN
Encryption	No — traffic is plain text	Yes — full encrypted tunnel
Anonymity	IP masking only	IP masking + encryption
Scope	Per-application config	All network traffic automatically
Performance	Faster (no encryption overhead)	Slower (encrypt/decrypt every packet)
Caching	Yes	No
Best for	Content bypass, caching, filtering	Secure communications, privacy

Proxy (Reverse) vs Load Balancer

Aspect	Reverse Proxy	Load Balancer
Does load balancing	Yes — one of its features	Yes — its only role
Hides server IPs	Yes	No
Caching	Yes	No
Logging & monitoring	Yes	No
Works with 1 server	Yes (for caching/security)	No point with 1 server
Access control	Yes	No

Proxy vs Firewall

Aspect	Proxy	Firewall
Layer	Application (L7) — understands HTTP	Network/Transport (L3-L4) — packet level
Inspects	Request content, URLs, headers	IP addresses, ports, protocols
Action	Allow/block by content or URL	Allow/block by IP/port rules
Caching	Yes	No
Anonymity	Yes	No
Combined?	Proxy firewall = both in one	Packet-level only by default

8. Complete Comparison Table

Proxy vs VPN vs Load Balancer vs Firewall

Aspect	Proxy	VPN	Load Balancer	Firewall
OSI Layer	Application (L7)	Network + App (L3-L7)	Network/App (L4-L7)	Network/Transport (L3-L4)
Encryption	No	Yes — full tunnel	No	No
Anonymity	Yes — IP masking	Yes + encryption	No	No
Caching	Yes	No	No	No
Load Balancing	Reverse proxy only	No	Yes — primary role	No
Access Control	Yes — URL/content based	Limited	No	Yes — packet/IP/port
DDoS protect	Reverse proxy — yes	Partial	No	Partial (rate limit)
Main purpose	Anon, cache, control	Secure comm. tunnel	Traffic distribution	Block malicious traffic

Proxy vs VPN vs Load Balancer vs Firewall across 8 dimensions

9. Quick Revision & Interview Questions

Quick Revision

-> Proxy = intermediary between client and server. Acts on behalf of one side.
-> Forward proxy: protects CLIENTS. Client → Proxy → Server. Server sees proxy IP only.
-> Reverse proxy: protects SERVERS. Internet → Proxy → Servers. Client sees proxy IP only.
-> Forward proxy uses: anonymity, bypass geo-blocks, content filtering, caching, access control.
-> Reverse proxy uses: server anonymity, DDoS protection, load balancing, CDN, SSL termination.
-> Proxy vs VPN: proxy hides IP only. VPN encrypts ALL traffic in a tunnel. VPN = more secure.
-> Proxy works per-application (App Layer). VPN works for all network traffic (Network Layer).
-> Reverse proxy vs Load Balancer: reverse proxy does everything LB does + caching, hiding, logging.
-> Load balancer = traffic distribution ONLY. Reverse proxy = full feature set including LB.
-> Proxy vs Firewall: proxy = application layer, content-aware. Firewall = packet-level, IP/port rules.
-> Proxy firewall = combination of both — content + packet filtering.
-> CDN is a reverse proxy: caches content geographically close to users to reduce latency.
-> NGINX, Cloudflare, AWS ALB, API Gateway — all are reverse proxies in practice.
-> DDoS: attackers hit reverse proxy — backend servers remain safe and hidden.

Interview Questions

■ 1. What is a proxy server? How is it different from a VPN?
■ 2. Explain the difference between a forward proxy and a reverse proxy.
■ 3. What is a reverse proxy used for? Give three real-world examples.
■ 4. Can a reverse proxy replace a load balancer? Can a load balancer replace a reverse proxy?
■ 5. How does a proxy protect against DDoS attacks?
■ 6. What is the difference between a proxy and a firewall?
■ 7. What is a CDN and how is it related to reverse proxies?
■ 8. Why does a forward proxy work per-application but a VPN works for all traffic?
■ 9. What happens when multiple clients request the same resource via a forward proxy?
■ 10. Design a secure web architecture for a high-traffic API using proxy concepts.

Key Terms

Proxy Server	Intermediary between client and server — forwards requests on behalf of one side
Forward Proxy	Sits between clients and internet — hides client identity from servers
Reverse Proxy	Sits in front of servers — hides server identity from internet clients
VPN	Encrypted network tunnel — secures ALL traffic, not just IP masking
Load Balancer	Distributes traffic across multiple servers — no caching or anonymity
Firewall	Packet-level filtering by IP/port/protocol — not content-aware
Proxy Firewall	Combination — content-aware + packet filtering
CDN	Content Delivery Network — reverse proxy caching content near users globally
SSL Termination	Reverse proxy handles HTTPS, forwards plain HTTP to backends
DDoS Protection	Reverse proxy absorbs attack traffic — backends remain shielded
Caching (Proxy)	Store frequently requested content to serve without hitting origin server
Application Layer	OSI Layer 7 — proxies operate here, understand HTTP/HTTPS content
IP Masking	Hiding real IP address behind proxy or VPN IP
Anonymity	Forward proxy hides client IPs; reverse proxy hides server IPs